

Blockchain and Application Stack

L3 Data

PostgreSQL 16 (20
entities, 3NF)
Redis 7 · IPFS

L4 API and Services

Express / Node 20 ·
FastAPI
WebSocket · EIP-
712 · MCP
middleware

L5 Presentation

React 18 + Vite
ERC-4337 AA ·
MetaMask / email



L1 Settlement / Network

Polygon zkEVM
Cardona (ZK proofs)
Ethereum Sepolia ·
CCIP · The Graph ·
Tenderly

L2 Smart-Contract Platform

Solidity 0.8.20 ·
OpenZeppelin v5
UUPS · TimeLock ·
RBAC